Fault-free vertex-pancyclicity in faulty augmented cubes

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Abstract

The n-dimensional augmented cube, denoted as AQ_n, a variation of the hypercube, possesses some properties superior to those of the hypercube. In this paper, we show that every vertex in AQ_n lies on a fault-free cycle of every length from 3 to 2^n, even if there are up to $n-1$ link faults. We also show that our result is optimal.

1 Introduction

A *graph G* is a triple consisting of a *vertex set* $V(G)$, an *edge set* $E(G)$, and a relation that associates with each edge two vertices called its endpoints [22]. We usually use a graph to represent the topology of an interconnection network (network for short). The hypercube is one of the most versatile and efficient interconnection networks discovered to date for parallel computation. The hypercube is ideally suited to both special-purpose and general-purpose tasks, and can efficiently simulate many other same sized networks [15]. We usually use Q_n to denote an n-dimensional hypercube. Many variants of the hypercube have been proposed. The augmented cube, recently proposed by Choudum and Sunitha [5], is one of such variations. An n-dimensional augmented cube AQ_n can be formed as an extension of Q_n by adding some links. For any positive integer n, AQ_n is a vertex transitive, $(2n-1)$-regular, and $(2n-1)$-connected graph with 2^n vertices. AQ_n retains all favorable properties of Q_n since $Q_n \subset AQ_n$. Moreover, AQ_n possesses some embedding properties that Q_n does not. Previous works relating to the augmented hypercube can be found in [2], [5], [11], [12], [13], [16], [17], [20], [21].

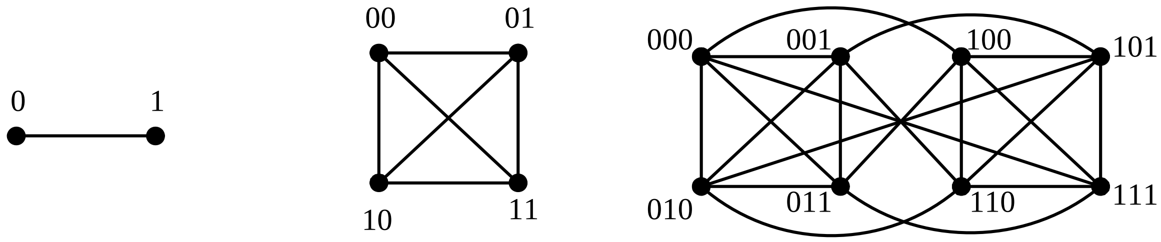
Linear arrays and rings, two of the most fundamental networks for parallel and distributed computation, are suitable for developing simple algorithms with low communication costs. Many efficient algorithms designed based on linear arrays and rings for solving a variety of algebraic problems and graph problems can be found in [15]. The *pancyclicity* of a network represents its power of embedding rings of all possible lengths. A graph G is called *pancyclic* whenever G contains a cycle of each length l for $3 \le l \le |V(G)|$. The arrangement graph [7], the hypercomplete network [6], the WK-recursive network [8], the alternating group graph [14], and the hyper-de Bruijn networks [9] are all pancyclic. A graph G is *vertex-pancyclic* (respectively, *edge-pancyclic*) if every vertex (respectively, edge) lies on a cycle of every length from 3 to $|V(G)|$. It is clear that if a graph G is edge-pancyclic, then it is vertex-pancyclic. The recursive circulant graphs with some condition [1], the alternating group graph [3], and the augmented cube [17] are edge-pancyclic.

Since faults may occur to networks, the fault tolerance of networks is an important issue in designing network topologies. Let $F_e \subset E(G)$ (respectively, $F_v \subset V(G)$) denote the faulty edges (respectively, the faulty vertices) in a graph G and let $F = F_e \cup F_v$. Suppose that $G - F$ is P, where P is pancyclic, vertex-pancyclic, or edge-pancyclic. Then, we call G $|F|$ fault-tolerant P. In addition, G is $|F|$-edge fault-tolerant P (respectively, $|F|$-vertex fault-tolerant P) if $F = F_e$ (respectively, if $F = F_v$). Note that if G is $|F|$ fault-tolerant P, then G is $|F|$-edge fault-tolerant P and $|F|$-vertex fault-tolerant P. Previously, the pancyclicity on various faulty networks was studied in [4], [10], [17], [18], [19], [20]. In [20], AQ_n has been shown to be $(2n-3)$ fault-tolerant pancyclic, where $n \ge 4$. Up to now, there is no research to concern fault-tolerant vertex-pancyclicity or fault-tolerant edge-pancyclicity. In this paper, we show that $AQ_n - F_e$ is vertex-pancyclic if $|F_e| \le n-1$, where $n \ge 2$. That is, we show that AQ_n is $(n-1)$-edge fault-tolerant vertex-pancyclic, where $n \ge 2$. In addition, we also show that our result is optimal.

The rest of this paper is organized as follows. In Section 2, the structure of the augmented hypercube is elaborated, and some definitions, notations, and properties used throughout this paper are introduced. In Section 3, we show that AQ_n is $(n-1)$-edge fault-tolerant vertex-pancyclic. In Section 4, this paper concludes with some remarks.

2 Preliminaries

Let G be a graph and let $u, v \in V(G)$. The degree of vertex v in G, written as $\deg_G(v)$, is the number of edges incident to v in G. In addition, $\delta(G) = \min\{\deg_G(v) | v \in V(G)\}$. A *path* $P[x_0, x_t] = \langle x_0, x_1, \cdots, x_t \rangle$ is a sequence of nodes such that two consecutive nodes are adjacent. t is the distance between nodes x_0 and x_t if $P[x_0, x_t]$ is a shortest path in G. We use $d_G(x_0, x_t)$ to denote the distance between x_0 and x_t in G, and use (u, v) to denote an edge whose endpoints are u



(a) (b) (c)

Figure 1. The structures of (a) AQ_1, (b) AQ_2, and (c) AQ_3

and v. Moreover, a path $\langle x_0, x_1, \cdots, x_t\rangle$ may contain other subpaths, denoted as $\langle x_0, x_1, \cdots, x_i, P[x_i, x_j], x_j, \cdots, x_t\rangle$, where $P[x_i, x_j] = \langle x_i, x_{i+1}, \cdots, x_{j-1}, x_j\rangle$. A *cycle* is a path with $x_0 = x_t$ and $t \geq 3$. A cycle (respectively, path) in G is called a *Hamiltonian cycle* (respectively, *Hamiltonian path*) if it contains every vertex of G exactly once.

An n-dimensional hypercube (n-cube for short) Q_n is an undirected graph with 2^n nodes each labeled with a distinct binary string $b_1b_2\ldots b_n$. Nodes $b = b_1\ldots b_i\ldots b_n$ and $b^i = b_1\ldots \overline{b_i} \ldots b_n$ are joined by an edge along dimension i, where $1 \leq i \leq n$ and $\overline{b_i}$ represents the one complement of b_i.

An n-dimensional augmented hypercube AQ_n is Q_n augmented by adding more links among its nodes (thus, $V(AQ_n) = V(Q_n)$). For a node $b = b_1b_2\ldots b_n$, it has $n - 1$ more links to connect to nodes $b_a^i = b_1b_2...b_{i-1}\overline{b_i}\,\overline{b_{i+1}}...\overline{b_n}$ in addition to its original n links, for all $i \in \{1, 2, \ldots, n - 1\}$. AQ_n has $2^{n-1}(n - 1)$ more links than Q_n. Note that AQ_1 is isomorphic to Q_1. Let AQ_{n-1}^0 (respectively, AQ_{n-1}^1) be the subgraph of AQ_n induced by $\{0b_2b_3\ldots b_n|\ b_i = 0 \text{ or } 1 \text{ for } 2 \leq i \leq n\}$ (respectively, $\{1b_2b_3\ldots b_n \mid b_i = 0 \text{ or } 1 \text{ for } 2 \leq i \leq n\}$). It is easy to see that AQ_{n-1}^0 (respectively, AQ_{n-1}^1) is isomorphic to AQ_{n-1}. In addition, AQ_n can be recursively constructed by adding 2^n edges between AQ_{n-1}^0 and AQ_{n-1}^1. A vertex $b = 0b_2b_3\ldots b_n \in V(AQ_{n-1}^0)$ is jointed to two vertices in AQ_{n-1}^1, which are $b^h = b^1 = 1b_2b_3\ldots b_n$ and $b^a = b_a^1 = 1\overline{b_2}\,\overline{b_3}...\overline{b_n}$. It is easy to see that an edge joins nodes b^h and b^a. The structures of AQ_1, AQ_2, and AQ_3 are shown in Figure 1. It is known that AQ_n is a vertex-transitive and $(2n - 1)$-regular graph [1]. The following three lemmas are needed to derive our main result.

Lemma 1. [17] Let $u, v \in V(AQ_n)$, where $n \geq 2$. There exists a path $P[u, v]$ of length l in AQ_n, where $d_{AQ_n}(u,v) \leq l \leq 2^n - 1$.

Lemma 2. [17] Let $u, v \in V(AQ_n)$, where $n \geq 2$. There exists a Hamiltonian path $P[u, v]$ in $AQ_n - F$, where $F \subset E(AQ_n)$, if $|F| \leq 2n - 4$. Moreover, there exists a cycle of length l in $AQ_n - F$ if $|F| \leq 2n - 3$, where $3 \leq l \leq 2^n$.

Lemma 3. Let $(x, y) \in E(AQ_{n-1}^0)$. Then (x^h, y^h) and (x^a, y^a) are also edges in AQ_{n-1}^1.

Proof. It is trivial if $y = x^i$, where $i \in \{2, 3, \ldots, n\}$. Now, consider that $y = x_a^i$ for some $i \in \{2, 3, \ldots, n - 1\}$. Suppose that $x = 0x_2x_3\ldots x_n$. Then, $y = 0x_2x_3...\overline{x_i}\,\overline{x_{i+1}}...\overline{x_n}$, $x^h = 1x_2x_3\ldots x_n$, $y^h = 1x_2x_3...\overline{x_i}\,\overline{x_{i+1}}...\overline{x_n}$, $x^a = 1\overline{x_2}\,\overline{x_3}...\overline{x_n}$, $y^a = 1\overline{x_2}\,\overline{x_3}...\overline{x_{i-1}}x_i...x_n$. Thus, $y^h = (x^h)_a^i$ and $y^a = (x^a)_a^i$. The result follows. □

3 Fault-free vertex-pancyclicity

In this section, we show that AQ_n is $(n - 1)$-edge fault-tolerant vertex-pancyclic. We format the theorem as follows.

Theorem 1. Let $F \subset E(AQ_n)$ denote the faulty edge set of AQ_n, where $n \geq 2$. $AQ_n - F$ is vertex-pancyclic if $|F| \leq n - 1$.

Proof. We proceed by induction on n. It is very easy to see that the theorem holds for AQ_2. As per our induction hypothesis, assume that the result holds for AQ_{n-1} for some $n \geq 3$. Consider that AQ_n and $F \subset E(AQ_n)$, where $n \geq 3$ and $|F| \leq n - 1$. For simplicity, we may assume $|F| = n - 1$. In addition, since AQ_n is vertex-symmetric, we only need to show that $z = 0^n$ (n consecutive 0's) lies on a cycle of length l in $AQ_n - F$, where $3 \leq l \leq 2^n$. Let $F_0 = F \cap E(AQ_{n-1}^0)$, $F_1 = F \cap E(AQ_{n-1}^1)$, and $F_c = F \cap (\{(b, b^h)|\ b \in V(AQ_{n-1}^0)\} \cup \{(b, b^a)|\ b \in V(AQ_{n-1}^0)\})$. The following cases are considered:

Case 1: $|F_0| = n - 1$. Thus, $|F_1| = |F_c| = 0$. Two cases are further considered:

Case 1.1: $3 \leq l \leq 2^{n-1} + 1$. Remember that an edge joins z^h ($= 10^{n-1}$) and z^a ($= 1^n$); that is, $d_{AQ_{n-1}^1}(z^h, z^a) = 1$. By Lemma 1, there exists a path $P[z^h, z^a]$ of length l_1 in AQ_{n-1}^1, where $1 \leq l_1 \leq 2^{n-1} - 1$. Therefore, the desired

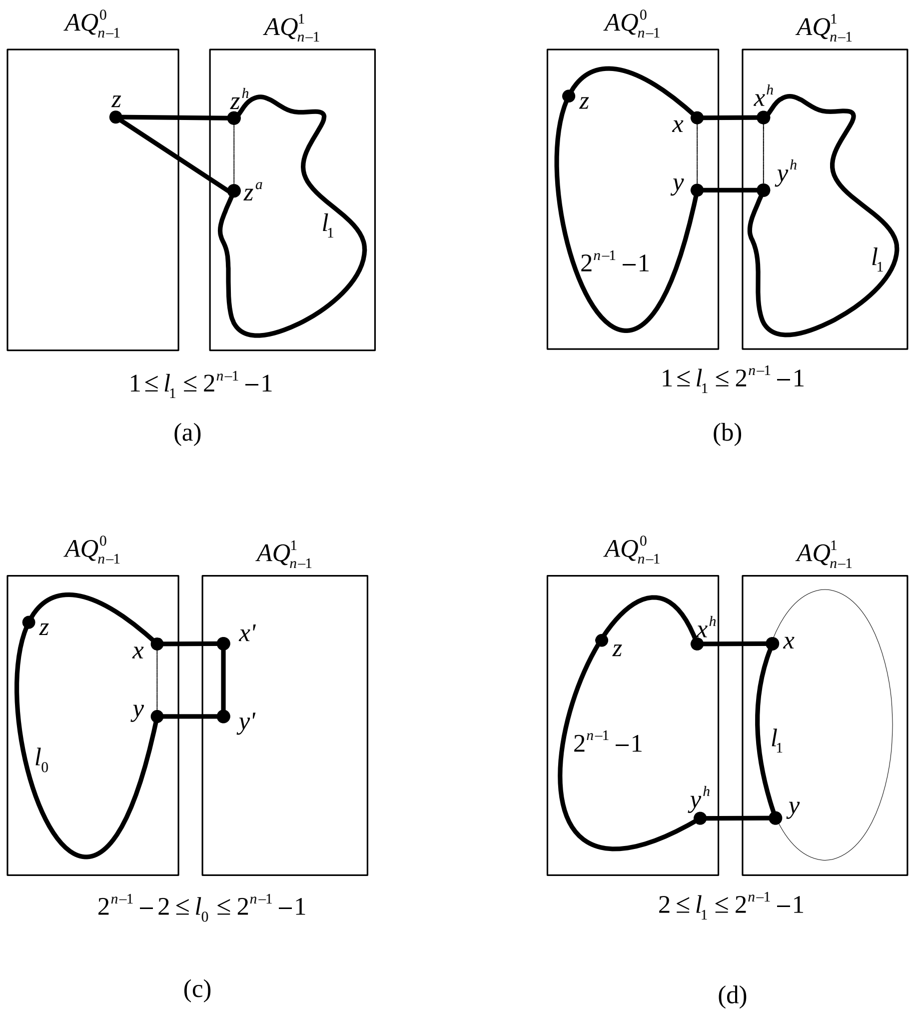


Figure 2. Construction of a cycle of length $l \in \{3, 4, \ldots, 2^n\}$ in $AQ_n - F$ with $F \subset E(AQ_n)$ and $|F| = n - 1$ (Theorem 1).

cycle of length $l = l_1 + 2 \in \{3, 4, \ldots, 2^{n-1} + 1\}$ can be constructed by $\langle z, z^h, P[z^h, z^a], z^a, z\rangle$ (see Figure 2(a)).

Case 1.2: $2^{n-1} + 2 \le l \le 2^n$. Let $(x', y') \in F_0$, we have $|F_0 - \{(x', y')\}| = n - 2 \le 2n - 5$, when $n \ge 3$. By Lemma 2, there exists a cycle C_0 of a Hamiltonian cycle in $AQ_{n-1}^0 - (F_0 - \{(x', y')\})$. If $(x', y') \notin E(C_0)$, then randomly select an edge (x, y) from $E(C_0)$. If $(x', y') \in E(C_0)$, then let $(x, y) = (x', y')$. Next, let $P[x, y] = C_0 - \{(x, y)\}$. Clearly, $z \in P[x, y]$, the length of $P[x, y]$ is $2^{n-1} - 1$, and $d_{AQ_{n-1}^1}(y^h, x^h) = 1$. By Lemma 1, there exists a path $P[y^h, x^h]$ of length l_1 in AQ_{n-1}^1, where $1 \le l_1 \le 2^{n-1} - 1$. Therefore, the desired cycle of length $l = l_1 + (2^{n-1}-1) + 2 \in \{2^{n-1}+2, 2^{n-1}+3, \ldots, 2^n\}$ can be constructed by $\langle x, P[x, y], y, y^h, P[y^h, x^h], x^h, x\rangle$ (see Figure 2(b)).

Case 2: $|F_0| \le n - 2$. Two cases are further considered:

Case 2.1: $3 \le l \le 2^{n-1}$. By the induction hypothesis, there exists a cycle C of the require length $l \in \{3, 4, \ldots, 2^{n-1}\}$ in $AQ_{n-1}^0 - F_0$ such that $z \in V(C)$. Clearly, C is the desired cycle.

Case 2.2: $2^{n-1} + 1 \le l \le 2^{n-1} + 2$. By the induction hypothesis, there exists a cycle C_0 in $AQ_{n-1}^0 - F_0$ such that $z \in V(C_0)$, where $2^{n-1} - 1 \le |V(C_0)| \le 2^{n-1}$. Moreover, let $(x, y) \in E(C_0)$ such that $(x, x^h), (y, y^h), (x^h, y^h) \notin F$ or $(x, x^a), (y, y^a), (x^a, y^a) \notin F$ *. Note that (x^h, y^h) and (x^a, y^a) are also edges in AQ_{n-1}^1 (by Lemma 3). Additionally, let $P[x, y] = C_0 - \{(x, y)\}$ and let l_0 be the length of $P[x, y]$. (Thus, we have $2^{n-1} - 2 \le l_0 \le 2^{n-1} - 1$.) Let $(x', y') \in \{(x^h, y^h), (x^a, y^a))\}$ such that $(x, x'), (y, y'), (x', y') \in E(AQ_n) - F$. The desired cycle of length $l = l_0 + 3 \in \{2^{n-1} + 1, 2^{n-1} + 2\}$ can be constructed by $\langle x, P[x, y], y, y', x', x\rangle$ (see Figure 2(c)).

* Since $|E(C_0)| \ge 2^{n-1} - 1$, we have at least $2^{n-1} - 1$ choices. If such an edge does not exist, then $|F| > (2^{n-1} - 1) > n - 1$, which is a contradiction.

Case 2.3: $2^{n-1} + 3 \le l \le 2^n$. First, consider that $n = 3$. We have $l \in \{7, 8\}$, $|F| = 2$, and $|F_0| \le 1$. By Lemma 2 (since $|F| = 2$), there exists a cycle C of length 8) in $AQ_3 - F$; obviously, C contains z and C is the desired cycle of length 8. If $|F_1| \le 1$, then by Lemma 2 (since $|F_0| \le 1$) there exists a cycle C_0 of length 4 in AQ_2^0. Let $(x, y) \in E(C_0)$ such that (x, x^h), $(y, y^h) \notin F_c$ or (x, x^a), $(y, y^a) \notin F_c$ *. Then, let $x' \in \{x^a, x^h\}$ and $y' \in \{y^a, y^h\}$ such that (x, x'), $(y, y') \notin F_c$. In addition, we can construct a path $P[y', x']$ of length 2 in $AQ_2^1 - F_1$†. The desired cycle of length 7 can be constructed by $\langle x, P[x, y], y, y', P[y', x'], x', x\rangle$. If $|F_1| = 2$ (thus $|F_c| = |F_0| = 0$), then let $P[x, y]$ be a path of length 2 in $AQ_2^1 - F_1$‡. Since $|F_0| = 0$, by Lemma 2, there exists a Hamiltonian path $P[y^h, x^h]$ in $AQ_2^0 - F_0$. The desired cycle of length 7 can be constructed by $\langle x, P[x, y], y, y^h, P[y^h, x^h], x^h, x\rangle$.

Now consider that $n \ge 4$. Since $|F_1| \le |F| \le n - 1 \le 2(n - 1) - 3$, by Lemma 2, there exists a Hamiltonian cycle C_1 in $AQ_{n-1}^1 - F_1$. Let $P[x, y]$ be the subpath of C_1 with length l_1 such that (x, x^h), $(y, y^h) \notin F_c$, where $2 \le l_1 \le 2^{n-1} - 1$§. Since $|F_0| \le n - 2 \le 2(n - 1) - 4$, by Lemma 2, there exists a Hamiltonian path $P[y^h, x^h]$ in $AQ_{n-1}^0 - F_0$ (certainly, $z \in V(P[y^h, x^h])$). The desired cycle of length $l = 2^{n-1} - 1 + 2 + l_1 \in \{2^{n-1}+ 3, 2^{n-1}+ 4, \ldots, 2^n\}$ can be constructed by $\langle x, P[x, y], y, y^h, P[y^h, x^h], x^h, x\rangle$ (see Figure 2(d)). □

The result is optimal with respect to the number of edge faults tolerated since there are distributions of n edge faults over a AQ_n such that some vertex can not lie on a fault-free cycle of length three in the faulty AQ_n. Consider that a vertex $u = 0^n$ (n consecutive 0's). Suppose that $F = \{(u, u_a^i) | i \in \{1, 2, \ldots, n - 1\}\} \cup \{(u, u^n)\}$. Thus, $(u, u^i) \notin F$, for all $i \in \{1, 2, \ldots, n - 1\}$. Since u^i is not a neighbor of u^j, for all $i, j \in \{1, 2, \ldots, n - 1\}$ and $i \ne j$, u cannot lie on a fault-free cycle of length 3.

4 Discussion and conclusion

Linear arrays and rings, two of the most fundamental networks for parallel and distributed computation, are suitable for developing simple algorithms with low communication costs. The *pancyclicity* of a network represents its power of embedding rings of all possible lengths. In this paper, using inductive proofs, we showed that AQ_n is $(n - 1)$-edge fault-tolerant vertex-pancyclic, where $n \ge 2$. That is, every vertex of an AQ_n with at most $n - 1$ faulty edges lies on a fault-free cycle of every length from 3 to 2^n. In addition, we also showed that our result is optimal.

AQ_n is $(2n - 3)$ fault-tolerant pancyclic (thus, $(2n - 3)$-edge fault-tolerant pancyclic), where $n \ge 4$ [20]. We have shown that AQ_n is $(n - 1)$-edge fault-tolerant vertex-pancyclic but not n-edge fault-tolerant vertex-pancyclic. Therefore, AQ_n is not n fault-tolerant vertex-pancyclic and not n-edge fault-tolerant edge-pancyclic. A topic for further research is to explore the vertex-pancyclicity and/or edge-pancyclicity of augmented cubes in the presence of hybrid faults.

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* Since $|E(C_0)| = 4$, we have four choices. If such an edge does not exist, then $|F_c| \ge 4$, which is a contradiction.

† It is easy to verify that we can construct a path of length 2 between two arbitrary vertices in $AQ_2 - F$, where $F \subset E(AQ_2)$ and $|F| \le 1$.

‡ It is easy to verify that we can find a path of length 2 in $AQ_2 - F$, where $F \subset E(AQ_2)$ and $|F| = 2$.

§ Since $|E(C_1)| = 2^{n-1}$, we have at least $2^{n-1}/2$ ($=2^{n-2}$)choices. If such a path does not exist, then $|F| \ge 2^{n-2} > n - 1$ when $n \ge 4$, which is a contradiction.

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